Project Title: Project AMR-Guard

Tagline: *Secure Disposal. Stronger Medicine.*

Target Region: Maharashtra State (Phase 1)

1. Executive Summary

AMR-Guard is a low-friction, bilingual (Marathi/English) digital ecosystem designed to intercept antibiotic waste before it enters the environment. By removing the barriers of complex logins and high data usage, it turns every citizen into a "guardian" of public health.

2. Systematic Project Structure

Phase A: User Interaction (The "Input" Layer)

- **Snap-and-Report (Offline OCR):** A lightweight image recognition tool.

Users take a photo of the medicine strip. The app identifies the drug name/ expiry using on-device processing to save data.

- **Simple Reporting:** A "One-Tap" button to report disposal. No name or Aadhaar required—the app uses a unique **Device Token** to track contributions anonymously.
- **Location Intelligence:** Uses GPS to identify the user's Taluka/District to provide the most relevant local disposal instructions.

Phase B: The Logic Core (The "System" Layer)

- **Bilingual Interface:** A seamless toggle between **Marathi and English**.
- **The "Eco-Hazard" Algorithm:** Based on the drug type scanned, the app provides a specific "Hazard Score" (e.g., *“This antibiotic is highly*

persistent in water; please do not flush!”).

- **Cloud Sync:** Data collected offline is synced to a central dashboard when the user reaches a Wi-Fi or 4G zone.

Phase C: Fulfillment (The "Output" Layer)

- **Guardian Maps:** High-contrast maps showing the nearest **"Safe-Drop Pharmacies"** or Government Health Centers.
- **Disposal Verification:** A QR-code exchange system. The pharmacist scans the user’s app code upon receiving the waste, validating the disposal.
- **Knowledge Capsules:** 10-second voice notes in Marathi explaining AMR basics, designed for low-literacy or busy users.

3. Gamification: The "Social

Habit" Engine

To ensure the app isn't deleted after one use, we implement a **Progressive Reward Tier**:

- **Tiers of Guardianship:**
- **Level 1: Scout** (First disposal reported).
- **Level 2: Sentinel** (5 strips safely disposed).
- **Level 3: AMR-Guardian** (Consistent monthly disposal + 3 community referrals).
- **The "Maha-Leaderboard":** A friendly competition between cities (e.g., *"Nashik is currently leading Mumbai in safe disposal rates this week!"*).
- **AMR Hero Badges:** Shareable social media stickers (WhatsApp/Instagram) to spread awareness.
- **Redemption Points:** Points can be traded for small "Health Credits" at partner chemists (e.g., discounts on

hygiene products like soap or handwash).

4. Expected Outcomes & Impact

Outcome Area	Professional Target
Environmental	**Zero Leaching:** Ensuring zero antibiotic molecules from participating households reach the groundwater.
Public Health	**Extended Drug Life:** By reducing environmental exposure, we slow the birth of superbugs, keeping current antibiotics effective for longer.
Data & Policy	**The AMR Heatmap:** Providing health authorities with a real-time map of antibiotic "surplus" areas to prevent over-prescription.

AMR Safe Disposal I...

	over-prescription.
Logistical	**Closed-Loop System:** Creating the first verifiable "Take-Back" chain in Maharashtra that links the citizen directly to the incinerator.

5. Why "AMR-Guard" Wins the First Prize

- **Practicality:** It recognizes that Maharashtra has diverse connectivity; the **Offline-First** and **No-Login** approach ensures 100% accessibility.
- **Scalability:** It starts in one district (e.g., Chhatrapati Sambhajinagar or Pune) and can be cloned across India using the same codebase.
- **The "Human" Element:** By using gamification and Marathi voice notes, it moves AMR from a "scary scientific concept" to a "simple community duty."
- **Field-Readiness:** Unlike purely theoretical models, this connects to the existing pharmacy network, making it a **Field-Ready solution** as requested by the competition.